

HTML Forms & Input Elements Assignment

Forms • Inputs • Select • Textarea • Validation

Submission Deadline: Monday, 11:59 PM

Objective: Build realistic HTML forms by selecting appropriate controls, input types, attributes, labels, validation rules, and form structure.

Topics Covered

- <form> and form attributes taught in class
- <input> and input types taught in class
- Input attributes such as name, value, placeholder, required, readonly, disabled, checked, selected, maxlength, minlength, min, max, step, pattern, autocomplete and other attributes covered in class
- <label> and proper label-control association
- <select>, <option> and related attributes
- <textarea> and its attributes
- Previously learned HTML: headings, paragraphs, formatting, links, images, lists, tables, pre/code and blockquote

General Instructions

1. Create a complete HTML document for each problem.
2. Use meaningful labels for every user-editable form control.
3. Choose input types based on the actual information being collected.
4. Use validation-related attributes where the scenario requires them.
5. Use name attributes so submitted form data can be meaningfully identified.
6. Do not use CSS or JavaScript. Focus on HTML form structure and attributes.
7. Use previously learned elements where they improve the page, including headings, paragraphs, lists, tables, links and images.
8. Do not add controls simply to increase the number of inputs. Every field must have a clear purpose.

Assignment Problem 1: College Admission Application Form

Scenario: A college wants to replace its paper admission application with a structured HTML application form for prospective students.

Requirements

1. Create a page heading and introductory paragraph explaining the admission form.
2. Collect full name, date of birth, email, mobile number, password/application PIN, gender, category and address.
3. Use appropriate input types instead of using text for every field.
4. Create a course/program <select> with at least six options.
5. Include a single-choice field and a multiple-choice field where appropriate.
6. Collect previous qualification, percentage/CGPA, passing year and school/college name.
7. Use suitable min, max, step, minlength, maxlength or pattern attributes where meaningful.
8. Mark genuinely mandatory fields using required.
9. Include a date field with sensible constraints.
10. Include a file-upload field for an admission document if taught.
11. Add a textarea for permanent address.
12. Add a declaration/consent checkbox that must be selected before submission.
13. Provide Submit and Reset controls.
14. Use meaningful name and value attributes throughout.
15. Associate every label correctly with its form control.
16. Use previously learned headings, paragraphs, links and an image where appropriate.

Challenge: Choose validation and input types according to the actual data rather than applying identical attributes to every field.

Assignment Problem 2: Hospital / Clinic Appointment Booking Form

Scenario: A clinic wants an online appointment request form where patients provide contact information, appointment preferences, doctor selection and visit details.

Requirements

1. Create an Appointment Booking page with a short explanation.
2. Collect patient name, email, mobile number and date of birth.
3. Provide appropriate appointment date and time controls.
4. Create a doctor dropdown with at least five fictional doctors and specialties.
5. Create a visit-type choice such as New Patient, Follow-up, Consultation and Emergency.
6. Create a preferred communication method choice.
7. Add a textarea for reason for visit/additional information.
8. Include an optional patient ID field.
9. Use suitable required, minlength, maxlength, pattern, min and max attributes.
10. Include a checkbox for accepting appointment terms.
11. Add Submit and Reset controls.
12. Use labels and meaningful name/value attributes.
13. Add a clinic contact/emergency link.
14. Use a small previously learned table outside the form to display clinic timings.

Challenge: Decide which information is required, optional, single-choice, constrained or better represented as long-form text.

Assignment Problem 3: Online Course Enrollment & Student Registration

Scenario: An education platform wants a registration form for students enrolling in a technical course, including account, course, learning preference and background information.

Requirements

1. Create an introductory course enrollment section.
2. Collect student name, email, password, mobile number and date of birth.
3. Create a course dropdown with at least six courses.
4. Create a batch/session dropdown with multiple options.
5. Provide radio choices for Online, Offline or Hybrid learning.
6. Provide checkboxes for areas such as Java, Python, Web Development, Data Structures and Database.
7. Collect education level, college/institution and current semester/year.
8. Add a number field for programming experience with sensible min/max/step.
9. Add a URL field for GitHub/portfolio if taught.
10. Add a textarea for learning goals.
11. Use appropriate placeholder, required, minlength/maxlength, pattern and other taught attributes.
12. Include a confirmation checkbox.
13. Provide Submit and Reset controls.
14. Use meaningful name/value attributes and correctly associated labels.
15. Add a course information link and relevant image outside the form.

Challenge: Design the form so that account data, single-choice preferences, multiple-choice interests, numeric data, URLs and long-form goals are represented correctly.

Assignment Problem 4: Employee Leave Request Form

Scenario: A company wants employees to submit leave requests through an HTML form containing employee details, leave category, dates, reason and confirmation.

Requirements

1. Create an Employee Leave Request page.
2. Collect employee name, employee ID, email and department.
3. Make employee ID required and apply sensible constraints where possible.
4. Create a department dropdown with at least six departments.
5. Create leave-type choices such as Casual Leave, Sick Leave, Earned Leave, Work From Home and Other.
6. Create start and end date fields with appropriate constraints.
7. Add a number field for leave days with sensible min/max/step.
8. Create a radio choice for whether the employee is reachable during leave.
9. Add a textarea for the leave reason.
10. Add an alternate contact number.
11. Include an optional supporting-document upload if taught.
12. Add a declaration checkbox.
13. Use readonly, disabled or other taught attributes only where their behavior makes sense.
14. Use meaningful labels, name/value attributes and validation.
15. Include Submit and Reset controls.
16. Add a small table outside the form showing example leave categories and descriptions.

Challenge: Model a realistic workflow and decide which fields should be editable, required, optional, constrained or represented by a particular control.

Assignment Problem 5: FINAL CHALLENGE — E-Commerce Checkout & Delivery Form

Scenario: An online store needs an HTML-only checkout page. A customer has selected products and must now provide account, delivery, payment and order information.

Requirements

1. Create a Checkout page with a short introduction.
2. Use a previously learned table to display at least three selected products with Product, Quantity, Unit Price and Total.
3. Create customer information fields for name, email and mobile number.
4. Create delivery fields for house/street, city, state, postal code and country.
5. Use appropriate input types and constraints for email, telephone, postal code, number and other fields.
6. Create state/country dropdowns with multiple options.
7. Create delivery choices such as Standard, Express and Same Day.
8. Create payment choices such as Card, Cash on Delivery and UPI/Online Payment.
9. Provide appropriate payment fields such as card holder name, card number, expiry date and security code if those inputs were taught.
10. Create a billing-address option using a checkbox.
11. Create a textarea for delivery instructions.

12. Add an optional coupon/promo code field with a sensible maxlength or pattern where appropriate.
13. Use required, minlength, maxlength, min, max, step, pattern, placeholder, autocomplete and other taught attributes where they genuinely improve the form.
14. Use meaningful name/value attributes.
15. Include Terms & Conditions and order-confirmation checkboxes.
16. Provide Place Order and Reset controls.
17. Add a Continue Shopping link and a relevant product/store image.
18. Use previously learned headings, paragraphs, formatting, lists, links and tables where they improve the page.
19. Do not use CSS or JavaScript.

Challenge: This is the most comprehensive problem: combine form controls, validation, selects, textareas, checkboxes/radios, tables, links and images into one coherent checkout experience.

Form Quality Checklist

- Every form control has a clear, meaningful label.
- Input type matches the data being collected.
- Required fields use appropriate HTML validation.
- min/max/step and minlength/maxlength are used where appropriate.
- pattern is used only when a genuine format rule exists.
- name and value attributes are meaningful.
- Radio buttons in one choice group share the same name.
- Checkboxes represent independent selections or confirmations.
- select/option is used for predefined choices.
- textarea is used for genuinely longer text.
- Submit and reset controls are provided.
- No CSS or JavaScript is used.

Submission Reminder

Deadline: Monday, 11:59 PM. Complete and submit your work before the deadline.